Prosodic Differences in the Voices of Transgender and Cisgender Women: Self-Perception of Voice - An Auditory and Acoustic Analysis

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Summary: Introduction. The voice is an important parameter for identifying the speaker's gender. Transgender people seek to adapt their bodies to gender identity, and transgender women have greater difficulties in achieving vocal acceptance. In this context, the evaluation of the various parameters of the voice of transgender and cisgender women is essential to make it possible to propose appropriate intervention measures.

Objectives. To identify the differences in vocal characteristics of transgender and cisgender women.

Methods. An sectional study was conducted. The sample comprised 20 transgender women and 20 cisgender women who underwent evaluation of acoustic parameters, emotional prosody, self-perception, and perception of gender by lay listeners.

Results. The vocal characteristics of transgender and cisgender women differ in terms of the following parameters: f0, glottal noise excitation (GNE), vocal intensity, speech range profile (SRP), the first three formants of the vowel /a/, and in terms of emotional prosody, including duration and melodic contour. Higher values were mostly found in the cisgender population, except for noise level and vocal intensity. In addition, in most cases lay listeners identified the voices of transgender women as belonging to the male gender. There was a negative correlation between vocal dissatisfaction and f0 among transgender women.

Conclusions. Even though they perform vocal adjustments, the voices of transgender women are different from cisgender women in terms of acoustic parameters, vocal extension, and emotional prosody including duration and melodic contour. These differences have repercussions on the perception of gender by listeners.

Key Words: Voice— Transgender person— Acoustics— Speech therapy— Prosody.

INTRODUCTION

Some individuals express identities opposite to their biological sex. Their lives demand social and legal recognition of a gender different from that informed by sex and seek vocal and body transformations to help them feel they belong to this new group. These individuals identify themselves as being transgender.[1,2] Transgender women have difficulties in finding a voice appropriate to their gender identity, which is an important problem for these people.

Hormonal treatment has not proved effective in producing a transgender female voice, even if it is used for a long time. This is because no hormone alone raises the fundamental frequency, reduces vocal fold mass, or has a lasting effect on the pitch of the voice [3−6] − unlike transgender men, in whom testosterone increases vocal fold mass, producing a low-pitched voice.[5,6] For this reason, transgender women have more voice-related complaints than transgender men.[7]

In order to develop any intervention, it is necessary to understand that several parameters are involved in the perception of gender through the voice, including those that distinguish the female from the male pattern. These include acoustic correlates (fundamental frequency, intensity, and duration), resonance, articulation, emotional prosody and vocabulary.[8−10]

The voice is a significant factor in the perception of gender and non-conformity of the voice with personal identification can generate feelings of inadequacy and a possible negative psychosocial impact. Thus, it is necessary to develop studies that address the detailed analysis of the voices of transgender women compared with those of cisgender women. Comparative studies help understand more in depth the vocal nuances that differentiate the two groups. The purpose is to help these people acquire voices that meet their expectations. Hence, this study aims to verify differences in vocal characteristics of transgender and cisgender women.

METHODS

Data were collected at the voice laboratory of a public university. There was proximity between the laboratory and the space for transgender people located in a public hospital.

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This space provides a reference in the transsexualization process and offers comprehensive care in several health specialties.

Forty adults participated in the research, aged 18 to 60 years. Of these, 20 were transgender women, selected from the space for transgender people. The remaining 20 were cisgender women, were students and university professors from a public university, with ages, weights, and heights equivalent to those of women in the transgender group. It was important to pair the participants of the groups because the anatomical elements are identical between women, however, they do not always have the same physiological characteristics.[11,12]

Dialect was also controlled since an accent can change prosodic values. Prosody is affected by numerous variables and is strongly linked to issues associated with linguistic variation.[13–15]

A statistical calculation showed that to achieve a 99% power of reliability, 17 participants would be needed in each group. The calculation was performed using the statistical program MINITAB, version 17. Women who were dysphonic and/or had a history of diseases that affect the voice or who had undergone laryngeal surgery or who had undergone speech therapy with a focus on pitch elevation were excluded from the research.

For the selection of participants, the consent of the institution in which they were recruited as participants was obtained. Then, the women were contacted and those who made themselves available to participate signed the Free and Informed Consent Form (ICF) and answered a questionnaire. This covered questions used to characterize the sample and define the inclusion criteria (age, weight, height, naturalness, presence of dysphonia and/or history of diseases that affect the voice, use of hormones, smoking, and speech therapy). The vocally healthy participants performed detailed voice assessments individually at the Voice Laboratory.

Data collection lasted about 40 to 60 minutes with each participant as detailed below:

Acoustic evaluation of the voice

To obtain the vocal acoustic parameters, CTS Informática, Voxmetria, Vocalogramaand Fonoview software were used, in addition to the Praat software. To guarantee the quality of the vocal record, the participants were asked to sit comfortably in an upright posture and the recording of the voices took place in a quiet environment with a Karsect HT-2 headset microphone attached and maintained at a 45° angle at a distance of four centimeters from the labial commissure. A noise reduction plate, brand Andrea Pureaudio USB-AS Digital Audio Adapter, was also used to prevent interference caused by background sounds. The sampling rate was 44000 Hz, with 16 bits per sample. With the use of Voxmetria, each participant performed the sustained emission of the vowel / ε /, in addition to counting from 1 to 10. The voice samples were edited in the program itself, disregarding the beginning and end of the emission, which avoided the initial attack of the emission and use of reserve air at the end. F0, mean intensity, jitter, shimmer, GNE ratio (glottal-to-noise excitation), and noise mean values and standard deviation were obtained. With the Vocalograma software in the speech range profile option, each participant was asked to count from 20 to 30, at weak, normal, and strong intensity, to enable the analysis of the speech range profile (SRP) measurement, both in frequency and intensity modes. In the Fonoview software, the phrase "papai trouxe pipoca quente" belonging to CAPE-V[16] was recorded. It was then exported to the Praat software. Of this sentence, only the first vowel / a / was selected, based on the segmentation of the unstressed syllable of /papa /, located between the unvoiced bilabial phoneme. From this sample, values of the first, second, and third formants (F1, F2, and F3) were extracted.

Prosody evaluation

for prosody registration, the Fonoview software was used and each participant was asked to read the following declarative sentences pertaining to the MAC Battery emotional prosody test: "amanhã é sábado", "acabei de vir do médico" and "recebi uma carta de meu pai".[17] This instrument proved to be a useful clinical tool in examining linguistic and communicative skills related to discursive, pragmatic-inferential, lexical-semantic, and prosodic processing. For each sentence read, one participant intentionally used different emotions chosen by the researcher (sadness, joy, anger, and neutrality), in order to verify different melodic contours. The phrases of the MAC Battery and the emotions were random, in order to obtain greater spontaneity when executing them. In addition, they were displayed by the researcher through writing, so that she did not provide prosodic clues and exclude possible imitations of the participants. To guarantee the best quality of the recordings, each phrase was repeated three times by the participants, at random, according to the script, to allow the judge to choose from among the three. The criterion for this choice was defined by the selection of phrases with better representation of emotion, absence of noise, undue disfluencies, and other sound behaviors of the participants that would impair the quality of the recording and the rhythm of speech and which could interfere in the melodic contour of the sentences. In addition, the fundamental frequency curve in the representation of emotions was analyzed, from the perspective of the acoustics, complete in Praat.

The voices recorded in the CTS Fonoview program were imported into the Praat program. The software allows the selection of each speech segment, which can be a phoneme, syllable, word, or phrase from the visualization of the spectrogram and the recorded audio. When selecting the segment, it is possible to obtain results from the parameters that were separated in each selected segment. To carry out the prosodic analysis, excerpts referring to each sentence were selected and values related to the duration of the

sentence emission and maximum, minimum, and mean f0 were extracted.

Evaluation of gender perception and emotion

the CTS Fonoview software was used to record vocal recordings, with a sampling rate of 44,000 Hz. To enable the evaluation of gender perception, the researcher asked each participant to read the six phonetically balanced phrases proposed in the CAPE-V protocol: "Érica tomou suco de pêra e amora", "Sônia sabe sambar sozinha", "olha lá o avião azul", "agora é hora de acabar", "minha mãe namorou um anjo" and "papai trouxe pipoca quente".[16] To assess the perception of emotion, the recordings belonging to the MAC emotional prosody test were used: "amanhã é sábado", "acabei de vir do médico" and "recebi uma carta de meu pai".[17]

For the trial, three lay judges were invited, without in-depth knowledge of vocal parameters or information about the objectives of and population involved in the study. In order to represent different social contexts, they were selected from the population of students from a public university, belonging to different centers (Center for Biological Sciences, Center for Engineering, Center for Philosophy and Human Sciences). The phrases were grouped by individual, randomized, containing 20% repetition (in order to analyze intra-rater reliability), and separated into two stages (gender perception and emotion perception). The audio samples collected were transferred to a CD and passed on to the evaluators.

In the first stage, the judges assessed gender perception, using the recordings of the six sentences of the CAPE-V protocol.[16] The judges identified the degree of femininity of the voice, answering if the voice heard was very feminine, not very feminine, neutral, not very masculine, or very masculine, according to the scale present in the voice questionnaire for transgender women - TWVQ.[18]

In the second stage, the judges evaluated the perception of emotion, using the recordings of the phrases of the MAC Battery.[17] However, only one sentence of each emotion was selected by the author and passed on to the evaluators, under the criterion of better representation of the emotion, observed by the auditory impression and melodic contour presented in the spectrogram. Therefore, the judges heard four phrases from each participant, one of anger, one of joy, one of sadness, and one of neutral, and answered what emotion the voice indicated.

Assessment of vocal self-perception for transgender participants

the TWVQ questionnaire was applied.[18] This seeks to measure the experiences of transgender women with their voices. It has 30 closed questions and the overall score ranges from 30 to 120 points, so the maximum score indicates vocal dissatisfaction. Each participant responded to items related to the social impacts of transgender vocal self-assessment. The answers to the questions varied between "never", "rarely" "sometimes", "almost always", and "always".[19] This step enabled the analysis of self-perception (the categorical variables).

The statistical measures of mean, standard deviation, and median were obtained for the numerical variables. For the comparison between the groups, involving categorical variables, Pearson's Chi-square or Fisher's Exact test was used; for equal variances, the Student's *t*test was used. The Student's t was used with unequal variances or Mann-Whitney, in the case of numerical variables. To assess the degree of the relationship between two numerical variables, Spearman's correlation coefficient and the specific Student's *t*test for the null correlation hypothesis were obtained. As for the degree of agreement between the judges, the Kappa coefficient was used. The verification of normality was performed using the Shapiro-Wilk test and the equality of variances was performed using the Levene F test.

The margin of error used in deciding the statistical tests was 5%. The data were entered into the EXCEL spreadsheet and the program used to obtain the statistical calculations was IMB SPSS version 23.0.

RESULTS

Most participants in the groups were between 21 and 29 years old (29.6 ± 8.55). In terms of Body Mass Index (BMI), just over half (55) were overweight in the transgender group, while this percentage was 35% in the cisgender group. 55% of the transgender women and 40% of the cisgender women were from the same city. However, currently, only one does not reside in the same state, but all were from the same geographic region of the country. As for the use of hormones, the majority (90.0%) of transgender women used them, whereas this percentage was zero among cisgenders. This statistic stands out because the process of corporal feminization carried out by transgender women includes hormone therapy.[20] Despite this, studies indicate that these medications do not alter vocal characteristics. Although these women have undergone hormonal treatment for a long time, they usually do not have vocal changes, as no hormone raises the fundamental frequency, reduces vocal fold mass, or has a lasting effect on the pitch.[21] There were no differences between the values of such variables ($P>0.05$) between the groups, which points to the balance in the sample composition.

Table 1 shows the normal/altered values and the F0 band, according to the obtained vocal acoustic data. Comparing the group of transgender women to the cisgender group, differences were observed in variable f0. The participants were divided into the following two groups: one with those whose F0 was between 150 and 250Hz and the other with those whose F0 was below 150Hz. For the other variables, the normal values described in the software itself were used. It was observed that in the group of transgender women, nine participants had an f0 below 150Hz. In the group of cisgender women, only one participant had f0 below 150Hz. For

TABLE 1.
Comparison of Normality Between Groups for Acoustic Measurements

Variable	Group						*P* value
	Transgender		Cisgender		Total Group		
	n	%	*n*	%	*n*	%	
TOTAL	20	100	20	100	40	100	
F0							$P^{\dagger}$ 0.003*
150 - 250Hz	11	55	19	95	30	75	
Below 150Hz	9	45	1	5	10	25	
Jitter							$P^{\ddagger}$= 1,000
Normal	18	90	19	95	37	92.5	
Altered	2	10	1	5	3	7.5	
Shimmer							$P^{\ddagger}$= 0.605
Normal	17	85	19	95	36	90	
Altered	3	15	1	5	4	10	
GNE							$P^{\ddagger}$= 0.487
Normal	18	90	20	100	38	95	
Altered	2	10	-	-	2	5	
Noise							$P^{\ddagger}$= 0.487
Normal	18	90	20	100	38	95	
Altered	2	10	-	-	2	5	

* Significant difference at the level of 5.0%.
† Using Pearson's chi-square test.
‡ Through Fisher's exact test.
GNE, glottal to noise excitation.

the other acoustic variables, there was no difference between the groups.

When the data were analyzed according to the mean/median of the groups (Table 2) differences for the following acoustic variables are verified: mean f0, SD f0, GNE proportion, and noise. The mean/median of such variables were higher in the cisgender group except for the noise, which was higher in the transgender group.

Table 3 describes the results of the first, second, and third formants obtained through the emission of the vowel /a/.

TABLE 2.
Statistics of Acoustic Parameters and Differences Between Groups

Variable		Group		*P* value
		Transgender (*n* = 20) Mean ± SD (Median)	Cisgender (*n* = 20) Mean ± SD (Median)	
Acoustic parameters	Intensity - Average	52.28 ± 6.18 (51.53)	50.88 ± 5.96 (51.25)	$P^{\dagger}$= 0.470
	F0 - Average	165.74 ± 44.40 (157.65)	195.17 ± 25.39 (199.16)	$P^{\ddagger}$= 0.015 *
	F0 - DP	1.59 ± 0.51 (1.44)	2.20 ± 1.35 (1.77)	$P^{\S}$= 0.021 *
	Jitter	0.30 ± 0.30 (0.19)	0.21 ± 0.12 (0.18)	$P^{\S}$= 0.645
	Shimmer	3.83 ± 2.38 (3.12)	3.51 ± 2.20 (2.63)	$P^{\S}$= 0.978
	GNE	0.75 ± 0.18 (0.78)	0.87 ± 0.08 (0.89)	$P^{\S}$= 0.036 *
	Noise	1.27 ± 0.76 (1.16)	0.77 ± 0.32 (0.71)	$P^{\S}$= 0.033 *

* Significant difference at the level of 5.0%.
† Using the *t* Student test with equal variances.
‡ Using the *t* Student test with unequal variances.
§ Using the Mann-Whitney test.
SD, standard deviation; F0, fundamental frequency; GNE, glottal to noise excitation.

TABLE 3.
Comparison of Vowel /a/ Formants Between Groups of Transgender and Cisgender Women

Formant	Group: Transgender ($n = 20$) Mean ± SD (Median)	Group: Cisgender ($n = 20$) Mean ± SD (Median)	*P* value
F1	728.90 ± 93.19 (725.11)	879.69 ± 95.94 (861.39)	$P^{\dagger}$<0.001 *
F2	1308.22 ± 114.79 (1279.08)	1437.51 ± 112.21 (1449.83)	$P^{\dagger}$= 0.001 *
F3	2411.74 ± 247.90 (2476.31)	2720.35 ± 181.86 (2666.65)	$P^{\dagger}$<0.001 *

* Significant difference at the level of 5.0%.
† Through the *t* Student test with equal variances.
SD, standard deviation; F0, fundamental frequency;GNE, glottal to noise excitation; F1, first formant; F2, second formant; F3, third formant.

The values of the three formants were lower in the group of transgender women, with differences registered between the groups.

Table 4 presents the values referring to the speech range profile (SRP). There were differences between the groups in the variables f0 minimum and f0 maximum in the weak and strong emission modes; minimum f0, maximum f0, vocal extension, and maximum intensity in normal emission mode. The maximum intensity variable in normal emission was higher in the group of transgender women.

TABLE 4.
Comparison of the Speech Range Profile (SRP) Between the Groups of Transgender and Cisgender Women

Variable	Parameter	Group: Transgender ($n = 20$) Mean ± SD (Median)	Group: Cisgender ($n = 20$) Mean ± SD (Median)	*P* value
SRP	Weak			
	F0 -Minimum	118.06 ± 27.43 (126.21)	145.44 ± 32.05 (152.93)	$P^{\S}$= 0.002 *
	F0 - Maximum	202.96 ± 49.91 (191.47)	253.76 ± 59.77 (243.29)	$P^{\S}$= 0.009 *
	F0 - Ext.	84.90 ± 45.44 (62.00)	104.27 ± 75.43 (83.29)	$P^{\S}$= 0.607
	Ext. ST	9.37 ± 4.50 (8.17)	9.43 ± 6.77 (7.64)	$P^{\S}$= 0.358
	Minimum Int.	42.64 ± 5.12 (40.33)	41.22 ± 1.49 (40.41)	$P^{\S}$= 0.968
	Maximum Int.	78.38 ± 6.64 (79.40)	77.08 ± 6.48 (78.02)	$P^{\dagger}$= 0.534
	Area	4.89 ± 1.94 (4.49)	4.71 ± 2.05 (4.18)	$P^{\S}$= 0.570
	Normal			
	F0 – Minimum	116.65 ± 27.23 (120.82)	144.32 ± 28.07 (145.96)	$P^{\S}$= 0.008 *
	F0 - Maximum	220.35 ± 68.13 (212.08)	273.33 ± 63.94 (258.42)	$P^{\S}$= 0.007 *
	F0 - Ext.	103.70 ± 57.50 (94.00)	129.01 ± 68.76 (110.62)	$P^{\S}$= 0.042 *
	Ext. ST	10.78 ± 4.26 (9.88)	11.00 ± 5.16 (9.72)	$P^{\S}$= 0.715
	Minimum Int.	41.42 ± 2.56 (40.45)	40.38 ± 0.47 (40.21)	$P^{\S}$= 0.107
	Maximum Int.	77.50 ± 8.98 (77.00)	71.52 ± 5.06 (71.74)	$P^{\dagger}$= 0.013 *
	Area	5.46 ± 1.90 (4.87)	4.79 ± 1.20 (4.97)	$P^{\S}$= 0.534
	Strong			
	F0 – Minimum	116.09 ± 22.25 (115.08)	145.39 ± 21.00 (149.60)	$P^{\S}$<0.001 *
	F0 - Maximum	229.99 ± 62.19 (215.92)	317.85 ± 90.46 (301.71)	$P^{\S}$= 0.001 *
	F0 - Ext.	113.90 ± 51.01 (96.89)	172.46 ± 102.01 (156.28)	$P^{\S}$= 0.051
	Ext. ST	11.61 ± 3.28 (11.62)	13.14 ± 6.16 (11.96)	$P^{\S}$= 0.705
	Minimum Int.	43.63 ± 5.94 (40.42)	40.31 ± 0.58 (40.21)	$P^{\S}$= 0.155
	Maximum Int.	84.10 ± 12.77 (81.26)	77.94 ± 5.57 (77.48)	$P^{\ddagger}$= 0.059
	Area	6.88 ± 2.81 (6.07)	6.95 ± 3.25 (6.16)	$P^{\S}$= 0.829

* Significant difference at the level of 5.0%.
† Using the *t* Student test with equal variances.
‡ The t- Student test with unequal variances.
§ Using the Mann-Whitney test.
SRP, speech range profile; SD, standard deviation; F0, fundamental frequency; ST, semitone; EXT, extension; INT, intensity.

TABLE 5.
Comparison of Mean/Median of Acoustic Parameters Related to Prosody (Emotions of Anger and Sadness) in Transgender and Cisgender Women

Variables	Group		*P* value
	Transgender (*n* = 20) Mean ± SD (Median)	Cisgender (*n* = 20) Mean ± SD (Median)	
RAGE			
Recebi uma carta de meu pai			
F0 – Average	155.41 ± 38.10 (155.44)	200.15 ± 28.11 (196.51)	$P^{\dagger}$<0.001 *
Duration	1.67 ± 0.25 (1.63)	1.76 ± 0.25 (1.73)	$P^{\ddagger}$= 0.160
F0 – Maximum	245.93 ± 110.37 (209.92)	260.74 ± 57.77 (248.70)	$P^{\ddagger}$= 0.123
F0 – Minimum	111.22 ± 23.39 (115.66)	124.31 ± 46.59 (113.18)	$P^{\S}$= 0.271
Amanhã é sábado			
F0 – Average	146.00 ± 28.43 (148.74)	195.84 ± 27.87 (196.38)	$P^{\dagger}$<0.001 *
Duration	1.17 ± 0.13 (1.17)	1.23 ± 0.14 (1.22)	$P^{\ddagger}$= 0.140
F0 – Maximum	211.55 ± 102.41 (185.82)	248.28 ± 54.65 (239.35)	$P^{\ddagger}$= 0.004 *
F0 – Minimum	107.67 ± 23.83 (111.64)	138.24 ± 50.62 (147.66)	$P^{\S}$= 0.021 *
Acabei de vir do médico			
F0 – Average	159.57 ± 31.20 (153.53)	206.57 ± 39.22 (203.29)	$P^{\dagger}$<0.001 *
Duration	1.48 ± 0.33 (1.43)	1.47 ± 0.20 (1.42)	$P^{\ddagger}$= 0.570
F0 – Maximum	200.36 ± 54.08 (186.79)	289.42 ± 69.93 (264.00)	$P^{\ddagger}$<0.001 *
F0 – Minimum	110.91 ± 27.10 (120.09)	143.90 ± 40.43 (156.50)	$P^{\ddagger}$= 0.004 *
SADNESS			
Recebi uma carta de meu pai			
F0 – Average	152.35 ± 33.74 (154.45)	192.62 ± 18.34 (193.40)	$P^{\dagger}$<0.001 *
Duration	1.69 ± 0.20 (1.61)	1.70 ± 0.18 (1.70)	$P^{\ddagger}$= 0.449
F0 – Maximum	230.21 ± 114.11 (199.65)	264.93 ± 48.49 (248.33)	$P^{\ddagger}$= 0.003 *
F0 – Minimum	110.82 ± 32.87 (109.95)	121.62 ± 44.78 (98.13)	$P^{\ddagger}$= 0.433
Amanhã é sábado			
F0 – Average	148.96 ± 31.47 (161.36)	179.46 ± 19.80 (172.23)	$P^{\ddagger}$= 0.003 *
Duration	1.14 ± 0.27 (1.07)	1.22 ± 0.18 (1.18)	$P^{\ddagger}$= 0.099
F0 – Maximum	213.27 ± 83.80 (201.60)	235.51 ± 36.65 (228.23)	$P^{\ddagger}$= 0.062
F0 – Minimum	110.96 ± 31.79 (108.59)	100.30 ± 37.01 (80.28)	$P^{\ddagger}$= 0.152
Acabei de vir do médico			
F0 – Average	152.59 ± 35.39 (154.49)	193.00 ± 21.48 (189.89)	$P^{\dagger}$<0.001 *
Duration	1.41 ± 0.16 (1.40)	1.52 ± 0.18 (1.52)	$P^{\S}$= 0.041 *
F0 – Maximum	196.95 ± 56.15 (186.67)	245.79 ± 32.95 (239.20)	$P^{\ddagger}$<0.001 *
F0 – Minimum	103.82 ± 27.41 (98.16)	124.81 ± 37.58 (124.96)	$P^{\ddagger}$= 0.055

* Significant difference at the level of 5.0%.
† Using the *t* Student test with equal variances.
‡ Using the Mann-Whitney test.
§ Through the *t* Student test with unequal variances.
SD, standard deviation; F0, fundamental frequency.

Tables 5 and 6 show the results of the comparison between the groups regarding the evaluation of prosody, using the emotions of anger, sadness, joy, and neutral, respectively. There is a difference between the groups in the mean fundamental frequency in all phrases and emotions. These values were higher in the group of cisgender women.

For the phrases spoken in anger, in addition to the average f0, there was a difference in the phrases "acabei de vir do médico" and "amanhã é sábado", regarding the maximum and minimum f0, with higher values in the cisgender group. No difference was identified in the duration variable for any of the phrases when comparing the groups.

The data obtained in the emotion of sadness show that in addition to the average f0, differences were found in the maximum f0 for the phrases "recebi uma carta de meu pai" and "acabei de vir do médico", being higher in the cisgender

TABLE 6.
Comparison of Mean/Median of Acoustic Parameters Related to Prosody (Emotions of Joy and Neutral) in Transgender and Cisgender Women

Variables	Group		*P* value
	Transgender ($n = 20$) Mean ± SD (Median)	Cisgender ($n = 20$) Mean ± SD (Median)	
HAPPINESS			
Recebi uma carta de meu pai			
F0 – Average	181.60 ± 38.02 (180.42)	239.88 ± 64.56 (251.30)	$P^{\dagger}$<0.001 *
Duration	1.75 ± 0.21 (1.72)	1.83 ± 0.27 (1.79)	$P^{\ddagger}$= 0.327
F0 – Maximum	262.04 ± 83.22 (260.86)	352.40 ± 55.93 (342.01)	$P^{\ddagger}$<0.001 *
F0 – Minimum	118.21 ± 28.20 (126.01)	140.37 ± 52.13 (136.29)	$P^{\S}$= 0.105
Amanhã é sábado			
F0 – Average	173.86 ± 36.52 (176.33)	250.02 ± 46.47 (257.15)	$P^{\ddagger}$<0.001 *
Duration	1.20 ± 0.18 (1.25)	1.36 ± 0.23 (1.27)	$P^{\dagger}$= 0.083
F0 – Maximum	251.64 ± 75.02 (249.38)	365.27 ± 77.20 (387.78)	$P^{\dagger}$<0.001 *
F0 – Minimum	119.79 ± 32.18 (126.99)	146.16 ± 38.17 (149.15)	$P^{\ddagger}$= 0.023 *
Acabei de vir do médico			
F0 – Average	179.18 ± 39.21 (184.26)	258.24 ± 37.64 (252.04)	$P^{\ddagger}$<0.001 *
Duration	1.48 ± 0.22 (1.44)	1.60 ± 0.18 (1.57)	$P^{\ddagger}$= 0.051
F0 – Maximum	241.12 ± 77.78 (237.92)	364.80 ± 65.26 (346.22)	$P^{\dagger}$<0.001 *
F0 – Minimum	127.72 ± 38.82 (138.32)	146.79 ± 36.45 (154.71)	$P^{\dagger}$= 0.079
NEUTRAL			
Recebi uma carta de meu pai			
F0 – Average	156.61 ± 34.45 (152.29)	196.38 ± 24.56 (194.84)	$P^{\dagger}$<0.001 *
Duration	1.44 ± 0.18 (1.41)	1.56 ± 0.18 (1.63)	$P^{\ddagger}$= 0.033 *
F0 – Maximum	216.08 ± 52.48 (204.33)	269.29 ± 51.18 (254.79)	$P^{\dagger}$= 0.002 *
F0 – Minimum	123.90 ± 43.97 (117.80)	111.55 ± 37.50 (97.16)	$P^{\ddagger}$= 0.417
Amanhã é sábado			
F0 – Average	153.18 ± 31.09 (151.20)	181.90 ± 21.01 (175.55)	$P^{\dagger}$= 0.001 *
Duration	1.05 ± 0.10 (1.04)	1.10 ± 0.13 (1.11)	$P^{\dagger}$= 0.245
F0 – Maximum	202.74 ± 49.08 (199.43)	250.63 ± 38.67 (250.35)	$P^{\ddagger}$= 0.002 *
F0 – Minimum	111.95 ± 33.13 (109.66)	106.12 ± 35.26 (88.11)	$P^{\ddagger}$= 0.705
Acabei de vir do médico			
F0 – Average	161.72 ± 38.21 (156.00)	194.76 ± 25.92 (193.61)	$P^{\dagger}$= 0.003 *
Duration	1.38 ± 0.22 (1.38)	1.39 ± 0.16 (1.39)	$P^{\dagger}$= 0.874
F0 – Maximum	243.84 ± 113.08 (198.16)	246.68 ± 41.59 (241.70)	$P^{\ddagger}$= 0.058
F0 – Minimum	118.42 ± 35.26 (120.79)	120.66 ± 52.16 (92.32)	$P^{\ddagger}$= 0.978

* Significant difference at the level of 5.0%.
† Using the *t* Student test with equal variances.
‡ Using the Mann-Whitney test.
§ Through the t- Student test with unequal variances.
SD, standard deviation; F0, fundamental frequency.

group. In this emotion, there was a difference in the duration variable only in the phrase "acabei de vir do médico". There was a longer time interval in the emission of phrases spoken by cisgender women. With regard to the minimum f0, there was no difference between groups in any sentence.

The results presented in Table 6 express the values of joyful and neutral emotions. The prosodic parameters of the emotion of joy express differences in the average f0 between the groups. This behavior was also observed in the maximum f0 for all phrases. For the variables where differences were recorded, it is noteworthy that the mean and median were correspondingly higher in the cisgender group.

The values related to neutral emotion showed differences between the groups in the variables of mean and maximum f0 in the phrases "recebi uma carta de meu pai" and "amanhã é sábado", both with lower mean and median in

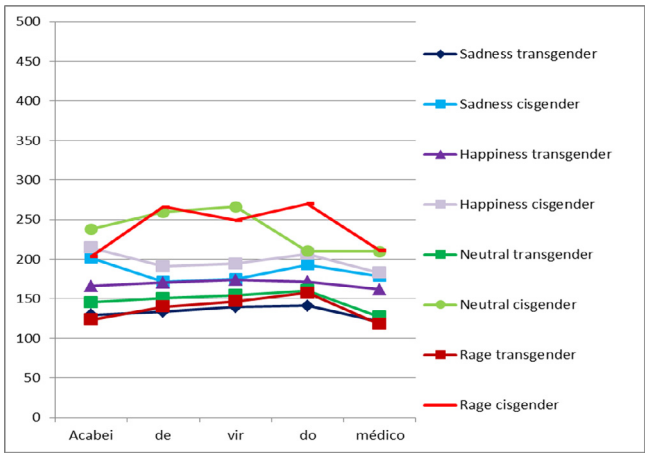

FIGURE 1. Stylized illustrative graphic of the intonation curves of the emotions of transgender and cisgender women.

the transgender group. However, in this first sentence, it was also possible to identify a difference in the sentence duration, which was produced with a longer time by cisgender women. The phrase "acabei de vir do médico" was not different between the groups in any variable, except for the mean f0.

Figure 1 consists of a graph showing the behavior of the intonation curves of the phrase "Acabei de vir do médico", in the emotions of sadness, joy, anger, and neutrality of an individual from the transgender group and an individual from the cisgender group. It was observed that the melodic curves are more accentuated in the cisgender participants, while the transgender participant had few variations and similar behaviors in the four emotions.

In the analysis of the vocal self-perception of transgender women using the TWVQ, the data indicate, in general, vocal dissatisfaction, with an average of 57 points in the final result of the questionnaire. Each participant could have a minimum score of 30 points and a maximum of 120 points. The questions that got the answer "always happens" by most volunteers were: "I am aware of how strangers perceive my voice", "I get upset when I am perceived as a man because of my voice" and "my voice makes me feel less feminine than I would like".

Regarding the characterization "almost always", it was common in questions that focused on the variability of tones in the speech. They were: "my spoken voice has a small variation of tones" and "the variation of tones of my female voice is small". Transgender women also reported with a greater consensus that sometimes situations happen in which the voice gets tired quickly, fails in the middle of the speech, and they get anxious when they know they need to use the voice. However, the questions that had the highest percentages of equal answers among transgender women were: "I get hoarse when I try to speak with my female voice", "my voice seems artificial", "I am less outgoing because of my voice", "I make a lot of effort to produce my voice ","when I laugh I look like a man ","I feel uncomfortable when talking to friends, neighbors or relatives because of my voice "and "my voice restricts the type of work I do". For these questions, the answers coincided with "never or rarely".

Regarding the correlation between the values of the TWVQ questionnaire and the fundamental frequency referring to the transgender group, a negative and significant correlation was found between the variables of average f0 and the TWVQ score (r = -0.620; P = 0.005). That is, the lower the f0 value, the greater the questionnaire's dissatisfaction values. It is worth noting that, in the TWVQ questionnaire, the highest responses indicate that negative situations of voice use occur more frequently in the daily lives of transgender people.

In assessing the degree of agreement between the pairs of the three judges, using the Kappa coefficient, it was observed, when analyzing each group separately, that for the transgender group, the scores varied from weak to good, with the majority being weak (0, 03 to 0.68). In the cisgender group, the judges moderately agreed, mostly, with six values of moderate agreement and one of good agreement, with scores ranging from -0.17 to 0.62.

Table 7 describes the assessment of the three judges regarding the degree of femininity/masculinity and the definition of emotion. It appears that 65% of the voices of the transgender group were defined as male, 15% neutral, 20% as female. In the cisgender group, 100% were defined as female voices. The other possible answers were not mentioned in this group by the judges. Differences were recorded between the two groups. As for the comparison of the perceptions of emotions, there was no difference between the groups (Table 7).

DISCUSSION

The results confirm the hypothesis that there are differences between transgender and cisgender women in several vocal parameters. The study of these parameters, such as fundamental frequency, intensity, speech range profile, formants, and prosody is important once they all interfere in the perception of gender and are crucial to the achievement of "passability" desired by transgender women.[22–24] The results of the comparison of acoustic parameters, considering the classifications of normal/altered values and the F0 band, show that only f0 was different between the two groups, with greater frequency in the transgender group. This was to be expected, regarding the greater length of the vocal folds of the participants of the transgender group.[25,26] Except for this parameter, the differences in the other variables were related to the presence of dysphonia.[27] Although there were some limited changes in the other values, the amount was small and similar in both groups. It is worth noting that no research participant had a history of dysphonia or vocal complaints. Even so, if you wish to have a normal larynx, you may experience changes in some acoustic parameters, because even without obtaining the vocal quality, you can implement supraglottic adjustments, in order to compensate for dysfunctional glottic conditions.[28,29]

TABLE 7.
Evaluation of the Degree of Femininity/Masculinity and Definition of Emotion in Transgender and Cisgender Women

Variable	Group						*P* value
	Transgender		Cisgender		Total Group		
	n	%	*n*	%	*n*	%	
TOTAL	20	100	20	100	40	100	
Degree of femininity / masculinity:							$P^{\dagger}$<0.001 *
Very feminine	3	15	14	70	17	42.5	
Little feminine	1	5	6	30	7	17.5	
Neutral	3	15	-	-	3	7.5	
Little male	8	40	-	-	8	20	
Very masculine	5	25	-	-	5	12.5	
Perception of emotion:							
RAGE							$P^{\ddagger}$= 0.057
Yes	8	40	14	70	22	55	
Not	12	60	6	30	18	45	
SADNESS							$P^{\ddagger}$ = 0.744
Yes	12	60	13	65	25	62.5	
Not	8	40	7	35	15	37.5	
HAPPINESS							$P^{\ddagger}$= 0.144
Yes	13	65	17	85	30	75	
Not	7	35	3	15	10	25	
NEUTRAL							$P^{\ddagger}$= 0.519
Yes	11	55	13	65	24	60	
Not	9	45	7	35	16	40	

* Significant difference at the level of 5.0%
† Using the Mann-Whitney test
‡ Through the test Chi-square Pearson(#) It was considered the fashion among the three examiners or the average in the case of ties.

With regard to the mean f0, of the 40 participants, 45% of transgender women and only 5% of the cisgender group had a voice frequency that was within the male range (below 150Hz). The discrepancy found between the two groups was expected due to anatomical conditions between the sexes.[22,30] It was also observed in a study carried out with 32 transgender women and 28 cisgender women that 3.5% of cisgender women and 34% transgender women were in the range of 80 to 150 Hz range of male voices.[31]

With regard to the numerical values of the acoustic variables, the mean f0, standard deviation of f0, GNE proportion, and noise differentiate the groups. However, only the noise level had a higher value in the transgender group. It is known that the presence of ~~(a)~~ voice disorders or possible supraglottic adjustments tends to alter the vocal signal in different ways, resulting in various types of noise in the vocal emission. Transgender women may make vocal adjustments in order to produce a more feminine voice. In addition, the combination of disturbance and noise measures improves the discrimination between signals with and without vocal deviation, suggesting that there is a higher level of noise. The voices of transgender women are probably more likely to have vocal deviation [29,32,33]

The f0 in cisgender women is known to range from 150 Hz to 250 Hz [34,35]whereas for transgender women there is still no consensual value in the literature. However, it has been pointed out that the f0 can vary from 116.8 Hz to 171 Hz.[28,36–40] The present research shows an average f0 of 165.74Hz for the study group (transgender) and 195.17Hz for the control group (cisgender). These results are consistent with what is recommended in the literature.[36–41] It is interesting to note, however, that the f0 in this study approaches the maximum value found in the surveys. This fact can be explained by the absence of vocal complaints in this population and by being a descriptive study. Intervention studies usually include participants with vocal dissatisfaction, which may be associated with lower f0. To support this statement, two studies stand out that also compare the f0 of the two groups without intervention, showing greater similarity with the data obtained in the present study.[31,42,43]

Regarding the standard deviation of f0, from a physiological point of view, it is directly linked to the vibratory

regularity of the vocal fold mucosa. Higher values of SD of f0 may indicate phonatory instability and irregularity of vocal fold vibration, causing a deviation in vocal production.[44,45] However, the standard deviation values for the cisgender group were slightly above the expected limit, since they should not exceed two Hz in individuals within the normal variability of vocal quality. Excessive f0 variability can be found in neurological vocal disorders or in situations of emotional stress.[46] The subjects' anxiety about the task may have caused this discreet deviation.

In contrast, transgender women had lower SD values of f0, suggesting better phonatory stability. The stability of the emission of a prolonged vowel is determined by the interrelation between the myoelastic forces of the larynx and the aerodynamics of the pulmonary current.[46] It is known that individuals with male physiology have longer and wider vocal folds, a smaller thyroid cartilage angle, longer vocal tracts, and larger lungs that result in higher values of subglottic pressure and airflow. Then, it can be inferred that they should show greater long-term phonatory stability, better supporting the emission which may justify the finding of the present study.[25,26,47,48]

The GNE proportion is the acoustic measure that calculates the noise produced by the vocal fold oscillation. Its values are considered normal when greater than or equal to 0.5dB. Lower GNE values may indicate inefficient glottic closure, greater additive noise in the voice, and a possible decrease in intensity.[44,49] In this study, the values for transgender women were significantly lower than those for cisgender women. However, it is noteworthy that they were mostly within the normal range. In fact, only two women in the transgender group had GNE changes. As for the comparison between the groups, differences were obtained, possibly due to the lower values of the group of transgender women.

The GNE measure is closely related to breathy vocal quality.[29,32,49] The breathy voice is usually evidenced by the escape of air between the vocal folds and its psychodynamics are associated with female sensuality.[28,46,50] Therefore, the results obtained in the present research suggest that transgender women may try to produce a more breathy voice in order to obtain the sensual aspect in the vocal quality.[28] In addition, when raising the pitch, they can enter the falsetto register causing the configuration of a parallel slit.[51]

The formants of the vowel / a /, in a pre-tonic position referring to the word "papai", were also analyzed. It is known that the acoustic characteristics, such as frequency of formants, amplitude, and duration, allow the vowels to be individually identified and these are related to the configurations of the articulators. In Brazilian Portuguese, as regards the production of the vowel / a /, it is classified as a low central vowel, not rounded and open. Regarding F1, it requires an adjustment of the jaw, which will influence the height of the tongue and is positioned lower in this vowel, resulting in the value of F1 being inversely proportional to the height of the tongue. F2 refers to the anteroposterior positioning of the tongue, which is more central in the vowel / a /. This value decreases when the tongue is positioned more posteriorly. F3 is related to the rounding of the lips, therefore, rounded vowels have lower F3 values compared to non-rounded ones.[52,53]

In this research, it was noticed that the values of F1, F2, and F3 were higher in the group of cisgender women. It is an indication that this group produced speech with greater lowering and anteriorization of the tongue and a more horizontal opening of the lips. As far as transgender women are concerned, they assume that they did not make such a wide mandibular excursion and, therefore, the back of the tongue must have been located closer to the hard palate. Based on formant issues, transgender women presented these articulatory configurations in a way that was more restricted than cisgender women.

The values of the first three formants may vary according to the population studied, as a sample of adult subjects from João Pessoa (PB), a city in northeastern Brazil, presented different measurements of the first (F1), second (F2), and third formants (F3) when compared with studies from other regions of the country.[54] In addition, gender can also be an influential factor in the results of formants, since female individuals have higher F1, F2, and F3 when compared to male individuals.[54,55]

In order to discuss the numerical results of the formants, reference values were adopted from a study conducted with cisgender men and women from João Pessoa, since accent patterns interfere in these results and all women participating in the present study also reside in this region. The vowel / a / numbers adopted as a reference for cisgender women were: 984Hz, 1589Hz, and 2831Hz for F1, F2, and F3 respectively. For cisgender men, they were: F1 728Hz, F2 1283Hz and F3 2534Hz.[56] When comparing the two surveys, it is clear that the F1 value of transgender women was equal to the F1 value found for male individuals. On the other hand, F2 was slightly higher than the group of cisgender men but well reduced when compared to the reference values of cisgender women. In F3, the values found in the study were lower than both groups in the reference study.

The cisgender women of the present study had values of F1, F2, and F3 slightly below the reference values for females and higher than those of males. In addition to the differences in the posture of oral structures in speech articulation, another factor that may have caused the difference between groups is related to anatomical and physiological characteristics typical of each sex, which imply specific acoustic parameters. This result is related to different typical dimensions of the vocal tract between female and male individuals.[22,30,57] Subjects with the shortest vocal tract have a higher frequency of formants, while in those with a longer vocal tract, formants tend to exhibit lower values. Therefore, biological males tend to have longer vocal tracts than biological females, regardless of the gender to which they identify.[58,59]

Regarding the speech range profile, it was noticed that the values of f0 extension in normal emissions and the values of minimum and maximum f0 in weak, normal, and strong emissions were higher in the group of cisgender women.

Their values were consistent with the literature, as it is known that the voice of transgender women is more severe and has lower f0 values than those of cisgender women, regardless of the intensity of production. Such an event is justified by the anatomical and vibratory differences of the vocal fold.[31,43]

The maximum intensity of normal emission differentiated the two groups, being higher in the group of transgender women. It can be inferred that, because the intensity results are equivalent in most of the tests performed, transgender women possibly make adjustments in the emission, in search of an approximation to the more feminine voice. One of the important parameters for the identification of the female gender is considered to be the presence of lower intensity. This result can also be related to the presence of breathiness in some women in the transgender group, contributing to the perception of a softer voice.[60,61]

The melodic contour of an utterance can undergo modifications within very wide limits depending on the mood of the speaker.[62] Thus, when comparing the group of transgender and cisgender women, it was noticed that the parameters that stood out as the most divergent were the average and maximum f0. Differences were obtained between the groups in all emotions and it was possible to notice that transgender women, in addition to presenting lower average, maximum and minimum frequencies than those in the control group, also had a lower range of melodic variation, resulting in phrases with a linear curve, possibly as a result of trying to maintain the desired f0.

The minimum f0 values were different in at least one sentence of the emotions of anger and joy. However, it was not consistent in all sentences and also inconsistent in the emissions of sadness and neutrality, bringing them closer to the values found in cisgender women. In contrast, the results obtained in the SRP showed differences between the groups regarding the minimum f0 in all emissions (strong, normal, and weak). This occurred because the group of cisgender women used a lower frequency in the emotional prosody test than in the SRP, since they were non-linear phrases in a melodic curve, unlike the count performed in the speech range profile. Transgender women, on the other hand, had very similar results in both tests. This finding suggests that in addition to a more linear speech, transgender women can have greater control in avoiding lowering the frequency to lower tones than in raising the frequency to the higher tones, also causing prosody with fewer variations when compared to cisgender women. Knowing that there are differences in characteristics between men and women regarding the vocal fold, such as length, mass, tension, vibration, stretching, subglottic pressure, larynx size, and lowering of the tract, it is understood that transgender women, even with vocal adjustments, will have more difficulties in reaching higher pitches and performing melodic variations with greater vocal extensions due to the anatomo-physiological characteristics of the vocal tract.[22,30,63]

When comparing the behavior of variables in emotions, it was noticed that cisgender women used an average f0 higher in emotions of joy and lower in emotions of sadness. The same behavior was repeated in the transgender group. However, when comparing angry and neutral emotions, cisgender women use higher frequencies in anger phrases and transgender women in neutral phrases. The literature confirms that the averages of the emission frequency values are higher in phrases that express the emotions of joy and anger, and lower in the expression of sadness, and intermediate in the expression of neutrality.[64] In this study, cisgender women exhibited this behavior; however, those in the transgender group showed lower emotions of anger than neutral ones. It is likely that in the realization of the emotion of anger, there was mandibular tension and this resulted in lower values of f0, unlike cisgender women. On the other hand, it is known that, in the expression of anger, there is an increase in the mandibular opening in most cases.[65] As the findings of the transgender population point to a lower amplitude of mandibular opening, inferred from the results of the second formant of the vowel / a /, this may have been responsible for the lower f0 numbers.

Regarding the perception of emotion, there were no differences between the groups. However, the judges obtained higher scores for agreement in the voices belonging to the cisgender group. This could indicate the presence of difficulty in the auditory perception of emotion, especially in the transgender group. Although the auditory perception of emotion was not statistically relevant when comparing the two groups, the acoustic and concordance data suggest that there are differences in the expression of emotions by transgender and cisgender women but are not perceived audibly by lay listeners. Nevertheless, future studies with more representative samples of judges are important to reinforce such results.

As for the assessment of gender perception by lay listeners, a large part of the transgender population was defined as having a male voice and no cisgender woman was included in this same group by the judges. It is known that transgender women, even with f0 within the expected range for women, are still perceived by their biological sex, pointing to the fact that other parameters of communication interfere in this judgment.[21,66] Considering that several measures were different between the groups of transgender and cisgender women, this can reinforce the view that the vowel formants, the speech range profile, intensity, and intonation may also be responsible for achieving the goal of passability of transgender women, not only at f0.

As for vocal self-perception applied only to transgender women, it was noticed that it had a relatively lower score when compared to the study by Schmidt (2018).[66] However, the women participating in the present study are part of a support group for the transgender population, being able to socialize among them and receive psychological counseling, with the main objective of promoting their acceptance. In spite of this, questions such as “I get upset when I am perceived as a man because of my voice” and “I am aware of how strangers perceive my voice” were encountered more often, which is compatible with the finding of the listeners'

perception of masculinity in the voice. The question with the lowest score was: "I get hoarse when I try to speak with my female voice" - which is also consistent with the fact that there are no individuals with vocal complaints and with the findings of few changes in acoustic parameters related to vocal quality.

In addition, 40% of the transgender women in the present study stated that they had a small spoken voice with varying tones, confirming the finding in the emotional prosody test and the f0 extension in normal emission. Likewise, 40% also stated that they perceive the tone of voice to be very important and feel less feminine than they would like because of the voice. Thus, the statement that transgender women have more restrained speech with a probable intention of controlling the frequency of the voice so as not to sound male supported since they also stated that they may not know how the tone of the voice will emerge when they start to speak.

When correlating the TWVQ data with the f0 average value, it was noticed that vocal dissatisfaction increases in cases with lower f0 values. Such an event occurs because vocal satisfaction is associated with having a more feminine voice, that is, with a higher f0, since it is a striking factor in the expression of gender, influencing, in different ways, interpersonal relationships and even professional work. When the voice is consistent with the expression of gender, social situations of speaking with or to other people becomes a more comfortable practice, without difficulties or dissatisfactions. The conformity between what is expressed and what the interlocutor sees does not generate any kind of discomfort, as there is no strangeness in the other as to the gender of the speaker. This conformity between what is expressed and what is seen socially is especially important in the sense of not embarrassing, exposing, or placing vulnerable situations of transgender people in the most diverse situations of social interaction and, consequently, influences the values of TWVQ.[7,42]

In view of the findings of this study, the importance of the parameters studied is used by speech therapists who perform vocal feminization procedures aimed at transgender women. The vocal parameters that differentiated the two groups of participants can form the basis for vocal therapy proposals, allowing greater satisfaction of women in relation to the use of the voice in different contexts of communication. In this way, they increase the means of achieving the goal of vocal "passability" desired by the transgender person. It also reinforces the idea that interventions focused only on adjusting the f0 may not produce the results expected by the therapist and the patient, since parameters of emotional prosody, formants, intensity, duration, and speech range profile, also cause vocal differences between transgender and cisgender women.

STUDY LIMITATIONS

One of the limitations of this study is the selection of transgender women. The study was carried out in a monitoring and support service for the gender transition, which could explain the large number of low scores in the TWVQ. In a population of trans women without specialized monitoring, the results will be different as they do not receive support for psychological and physiological issues involving transport.

Another important limitation was the small number of judges who participated in the auditory-perceptual analysis of gender end emotion. Only three judges took part in this study, although they belonged to different centers at the university. Future studies should encompass more judges to assess the voices, thus reinforcing the results obtained in this investigation.

Another limitation was the absence of a decibel meter to control the intensity, mainly in the assessment of the speech range profile. In order to be considered weak, habitual, or strong, they must be between 60-70dB, 70-80dB, and above 80dB respectively.

Performing the analysis of the formants only of the vowel / a / also presents itself as a limitation of the study, since the analysis of the other vowels could gather more information about the articulatory adjustments. However, in order not to extend the participants' evaluation and generate fatigue, it was decided to take advantage of the CAPE-v phrase, with the same context as that of the words used in the carrier phrases in a pre-tonic position, in the case of the vowel / a / of the word papai.

CONCLUSIONS

The vocal characteristics of transgender and cisgender women differ in terms of average, minimum, and maximum f0, GNE, noise, intensity, f0 extension, formants of the vowel /a/, and in emotional prosody including duration and melodic contour. In addition, lay listeners can identify the voices of transgender women as being male gender in most cases. Finally, there is a negative correlation between vocal dissatisfaction and the f0 of transgender women.

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